CivicVoice

A web-based platform bridging the gap between citizens and municipal authorities through transparent, efficient complaint resolution

Resolving Issues in Real-Time

PROJECT BY

Team Number: 30

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GUIDED BY

Dr. Sai Anirudh Karre

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Why Change is Urgent

Rapid urbanization is straining municipal services, leading to widespread dissatisfaction and an urgent need for more effective solutions.

The Growing Urban Crisis

Despite best efforts, citizens are often unheard, and critical issues remain unresolved for far too long. This lack of responsiveness erodes trust and hinders urban development.

2.5M+

Unresolved Complaints Annually

45+ Days

Average Resolution Time

32%

Citizen Satisfaction Rate

Current State

- Opaque processes
- Slow responses
- Low citizen engagement

Needed State

- Transparent tracking
- Rapid resolution
- High citizen trust

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Our Solution: Transparent & Connected

Public Interactive Map

All complaints visible to citizens, promoting transparency and collective awareness

Admin Dashboard

Streamlined verification, department routing, and resolution tracking workflows

Real-Time Updates

OTP login, NLP summarization, chatbot support, and SMS notifications at every stage

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System Architecture

01

Citizen Frontend

React-based interface for complaint submission with OTP authentication

02

Backend Services

Node.js/Express APIs handling NLP, department routing, and notifications

03

Database Layer

MongoDB Atlas storing complaints, status updates, and media

04

External Integration

Twilio, Nodemailer, and OpenStreetMap powering communication and mapping

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Technology Stack Powering CivicVoice

Frontend

- React.js with Vite for fast builds
- Tailwind CSS & Framer Motion
- Leaflet maps for visualization
- Axios for API calls

Backend

- Node.js & Express.js
- JWT authentication
- NLP module for summarization
- Cron jobs for escalation

Database & Services

- MongoDB Atlas cloud storage
- Twilio for OTP & SMS
- Nodemailer for emails
- OpenStreetMap tiles

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Citizen Journey: From Report to Resolution

Secure Login

OTP-based mobile authentication ensures verified access

Submit Complaint

Text, photos, videos with map-pinned location

Auto-Summarization

NLP generates concise summary for faster review

Track Status

Real-time updates from Pending to Verified to Resolved

Provide Feedback

Star ratings help improve service quality

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Admin Dashboard: Command & Control

Empowering Super Admin Ankita

Comprehensive Overview

Total complaints, active issues, resolution metrics at a glance

Smart Verification

Review NLP summaries alongside original complaints for quick decisions

Intelligent Routing

Auto-suggested departments streamline assignment workflow

Automated Notifications

Email to departments, SMS to citizens—no manual follow-up needed

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Key Features Driving Impact

Public Map Visibility

Every citizen sees all complaints, fostering community awareness and collective accountability

Intelligent Chatbot

Decision-tree bot helps classify complaints and guides users through the reporting process

Multi-Channel Alerts

SMS at verification and resolution, plus email notifications for department coordination

NLP Summarization

Machine learning condenses lengthy complaints into actionable summaries for efficient review

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Meet the Team Behind CivicVoice

Harsh Raj

System Architect

- MERN architecture design
- REST API implementation
- OTP authentication system
- Automated escalation logic

Chaitanya Prabhu

Integration Specialist

- Twilio SMS integration
- Email service setup
- Map service backend
- NLP module development

Lalithamsh

UX Designer

- High-fidelity mockups
- User flow diagrams
- Personas & patterns
- Interface design system

Prakash Bhabad

Frontend Developer

- React UI implementation
- Tailwind & animations
- API integration
- Complaint form & tracking

Deepti Koranga

Admin Interface Lead

- Dashboard analytics view
- Verification workflows
- Department assignment UI
- State management optimization

Repository: github.com/DeeptiKoranga/CivicVoice | Demo: youtu.be/XP3ObswnLxI

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Ready to Transform Civic Engagement

Get Started Today

Clone Repository

Download from GitHub and install dependencies

Configure Environment

Set up MongoDB, Twilio, and email credentials

Launch Platform

Run backend on port 5001, frontend on 5173

Local Setup: Access the full platform at localhost:5173 after following our simple installation guide

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Constraints & Limitations

Limited Budget

Development within tight financial resources requires efficient solutions and careful prioritization of features.

Time Constraints

Project timeline demands agile development and focused effort to meet ambitious launch deadlines.

Technical Limitations

Navigating existing platform compatibility and integration challenges for seamless system functionality.

Resource Availability

Ensuring sufficient human resources and infrastructure support for sustained growth and maintenance.

Scalability Challenges

Designing the system for future expansion and increased user load without compromising performance.

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Challenge & Solution

The Challenge

Outdated civic engagement systems lead to frustration, slow response times, and a significant disconnect between citizens and local governance.

Opaque Processes

Lack of transparency creates distrust and confusion among citizens.

Slow Resolution

Bureaucratic hurdles and manual workflows delay critical actions.

Low Citizen Trust

Inefficient systems erode public confidence in local governance.

Manual Workflows

Heavy reliance on paper and manual routing creates bottlenecks.

Our Solution

CivicVoice empowers communities with a transparent, efficient, and real-time platform, fostering accountability and rapid problem-solving.

Real-Time Transparency

Public map and status updates for every complaint build trust.

Rapid Response

AI-powered routing and automated notifications accelerate resolution.

Enhanced Trust

Citizen feedback loops and verified reporting build confidence.

Streamlined Operations

Digital workflows and admin dashboards boost efficiency.

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